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Department of Examinations – Sri Lanka

G.C.E. (A/L) Examination – 2024

02 - Chemistry

Marking Scheme

This has been prepared for the use of marking examiners. Some changes would be made according to the views presented at the Chief Examiners' meeting.

Amendments to be included

G.C.E. (A/L) Examination - 2024

02 - Chemistry

Distribution of Marks

Paper I	:	1 X 50	=	50
Paper II	:			
Part A	:	100 X 4	=	400
Part B	:	150 X 2	=	300
Part C	:	150 X 2	=	300
Total			=	1000
Paper II - Final Marks			=	100

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Common Techniques of Marking Answer Scripts.

It is compulsory to adhere to the following standard method in marking answer scripts and entering marks into the mark sheets.

1. Use a red color ball point pen for marking. (Only Chief/Additional Chief Examiner may use a mauve color pen.)
2. Note down Examiner's Code Number and initials on the front page of each answer script.
3. Write off any numerals written wrong with a clear single line and authenticate the alterations with Examiner's initials.
4. Write down marks of each subsection in a △ and write the final marks of each question as a rational number in a □ with the question number. Use the column assigned for Examiners to write down marks.

Example: **Question No. 03**

(i) ..... √ △ 4/5

(ii) ..... √ △ 3/5

(iii) ..... √ △ 3/5

03 (i) $\frac{4}{5}$ + (ii) $\frac{3}{5}$ + (iii) $\frac{3}{5}$ = $\boxed{\frac{10}{15}}$

MCQ answer scripts: (Template)

1. Marking templets for G.C.E.(A/L) and GIT examination will be provided by the Department of Examinations itself. Marking examiners bear the responsibility of using correctly prepared and certified templates.
2. Then, check the answer scripts carefully. If there are more than one or no answers Marked to a certain question write off the options with a line. Sometimes candidates may have erased an option marked previously and selected another option. In such occasions, if the erasure is not clear write off those options too.
3. Place the template on the answer script correctly. Mark the right answers with a 'V' and the wrong answers with a '0' against the options column. Write down the number of correct answers inside the cage given under each column. Then, add those numbers and write the number of correct answers in the relevant cage.

structured essay type and assay type answer scripts:

1. Cross off any pages left blank by candidates. Underline wrong or unsuitable answers. Show areas where marks can be offered with check marks.

2. Use the right margin of the overland paper to write down the marks.

3. Write down the marks given for each question against the question number in the relevant cage on the front page in two digits. Selection of questions should be in accordance with the instructions given in the question paper. Mark all answers and transfer the marks to the front page, and write off answers with lower marks if extra questions have been answered against instructions.

4. Add the total carefully and write in the relevant cage on the front page. Turn pages of answer script and add all the marks given for all answers again. Check whether that total tallies with the total marks written on the front page.

Preparation of Mark Sheets.

Except for the subjects with a single question paper, final marks of two papers will not be calculated within the evaluation board this time. Therefore, add separate mark sheets for each of the question paper. Write paper 01 marks in the paper 01 column of the mark sheet and write them in words too. Write paper II Marks in the paper II Column and wright the relevant details.

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Department of Examinations – Sri Lanka

අ.පො.ස.(උ.පෙළ)විභාගය/G.C.E. (A/L) - 2024

විෂයය අංකය
Subject No: 02

විෂයය
Subject: **Chemistry**

ලකුණු දීමේ පටිපාටිය/Marking Scheme

I පත්‍රය/Paper I

ප්‍රශ්න අංකය Question No.	පිළිතුරු අංකය Answer No.	ප්‍රශ්න අංකය Question No.	පිළිතුරු අංකය Answer No.	ප්‍රශ්න අංකය Question No.	පිළිතුරු අංකය Answer No.	ප්‍රශ්න අංකය Question No.	පිළිතුරු අංකය Answer No.	ප්‍රශ්න අංකය Question No.	පිළිතුරු අංකය Answer No.
01.	5	11.	3	21.	4	31.	4/5	41.	1
02.	2	12.	2	22.	3	32.	5	42.	1
03.	4	13.	3	23.	4	33.	3	43.	1
04.	4	14.	1	24.	1	34.	3	44.	1
05.	2	15.	3	25.	1	35.	3	45.	2
06.	4	16.	2	26.	4	36.	4	46.	3
07.	5	17.	4	27.	2	37.	5	47.	4
08.	3	18.	5	28.	5	38.	1	48.	3
09.	3	19.	5	29.	(all)	39.	2	49.	1
10.	5	20.	2	30.	5	40.	2	50.	1

✪විශේෂ උපදෙස්/Special Instructions:

එක් පිළිතුරකට ලකුණු 01 බැගින්/ 01 Mark for each question

මුළු ලකුණු/Total Marks 01 × 50 = 50

PART A — STRUCTURED ESSAY

*Answer **all four** questions on this paper itself. (Each question carries **100** marks.)*

1. (*a*) State whether the following statements are **true** or **false** on the dotted lines. Reasons **not** required.

(i) Particles in both cathode rays and β rays have the same charge to mass ratio (e/m). ..... true .....

(ii) In a copper (Cu) atom there are 6 electrons having magnetic quantum number, $m_l = -1$. ..... true .....

(iii) The F_2ClO^+ ion has a trigonal planar shape. ..... false .....

(iv) Sulphur (S) has the **lowest** first ionization energy among the elements F, S and Cl. ..... true .....

(v) Rules related to polarizing power of cations and polarizability of anions, predict that the melting point of LiCl is **higher** than that of KF. ..... false .....

(vi) In nitrous acid (HNO_2) the two N — O bonds are equal in length. ..... false .....

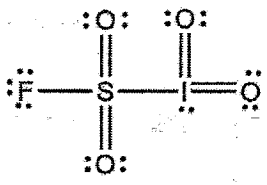
(vii) The number of Lewis dot-dash structures (resonance structures) that can be drawn for the ion CN_2^{2-} is 3. ..... true .....

(viii) The boiling point of hexane is **higher** than that of 2,2-dimethylbutane. ..... true .....

(04 marks X 8 = 32 marks)

1(a): 32 marks

(*b*) (i) Draw the **most** acceptable Lewis dot-dash structure for the molecule ISO_4F. Its skeleton is given below.



(06 marks)

(ii) Give oxidation numbers of the S and I atoms in the structure drawn in (i) above.

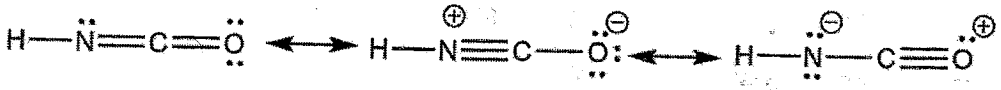
S ..... +5 or +V ..... I ..... +4 or +IV .....

(01) + (01)

Award marks for (ii) only of the Lewis structure in (i) is correct.

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(iii) An acceptable (stable) Lewis dot-dash structure for the HNCO molecule is given below. Draw **two** more Lewis dot-dash structures (resonance structures) for this molecule and indicate their stabilities relative to the structure given by writing **stable** or **less stable** or **unstable** under these structures.

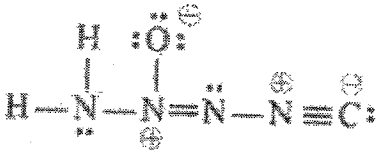


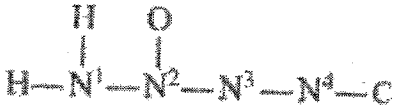
(less stable) (unstable)

(02 marks structure) + (01 mark stability)

(06 marks)

(iv) Complete the table based on the Lewis dot-dash structure and its labelled skeleton given below.





		N^1	N^2	N^3	N^4
I	The number of VSEPR pairs around the atom	4	3	3	2
II	electron pair geometry around the atom	tetrahedral	trigonal planar	trigonal planar	linear
III	Shape around the atom	pyramidal / trigonal pyramidal	trigonal planar	angular/ V / bent	linear
IV	Hybridization of the atom	sp^3	sp^2	sp^2	sp

(01 mark X 16 = 16 marks)

- Parts (v) to (viii) are based on the Lewis dot-dash structure given in part (iv) above. Labelling of atoms is as in part (iv).

(v) Identify atomic/hybrid orbitals involved in the formation of σ bonds between the two atoms given below.

I.	H	1s	N^1	sp^3
II.	N^1	sp^3	N^2	sp^2
III.	N^2	sp^2	O	$2p / sp^3$
IV.	N^2	sp^2	N^3	sp^2
V.	N^3	sp^2	N^4	sp
VI.	N^4	sp	C	2p / sp

(01 mark X 12 = 12 marks)

(vi) Identify the atomic orbitals involved in the formation of π bonds between the two atoms given below.

I.	N^2	2p	N^3	2p
II.	N^4	2p	C	2p
	N^4	2p	C	2p

(01 mark X 6 = 06 marks)

(vii) State approximate bond angles around the N^1, N^2, N^3 and N^4 atoms.

N^1	$(107° \pm 1)$	N^2	$(120° \pm 1)$
N^3	$(117° \pm 2)$	N^4	$(180° \pm 1)$

(01 mark X 4 = 04 marks)

(viii) Arrange the atoms N^1, N^2, N^3 and N^4 in **increasing** order of their electronegativities.

..... N^1 < ..... N^3 < ..... N^2 < ..... N^4

(04 marks)

1(b): 56 marks

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(*c*) Arrange the following species in the **increasing** order of the property indicated in parenthesis. (Reasons are **not** required)

(i) B, O, F, S, Na, Mg (electronegativity)

$Na < Mg < B < S < O < F$ **(06 marks)**

(ii) K^+, Mg^{2+}, Ca^{2+}, Al^{3+}, Cl^-, S^{2-} (ionic radius)

$Al^{3+} < Mg^{2+} < Ca^{2+} < K^+ < Cl^- < S^{2-}$ **(06 marks)**

1(c): 12 marks

2. (*a*) (i) I. **X** is an orange coloured ionic compound. It is composed of three elements in the ratio 7:2:2 (not in the order of writing the chemical formula). Of these, two are metals that belong to the same row of the Periodic Table. One of these two metals belongs to the *s*-block and the other to the *d*-block. The *d*-block metal is widely used for electroplating.

I. **X:** $K_2Cr_2O_7$

II. **Y** is a mineral acid. It is composed of three elements in the ratio 1:2:4 (not in the order of writing the chemical formula). One of the elements in **Y** is also found in **X**. **Y** is used in the manufacture of phosphate fertilizer.

II. **Y:** H_2SO_4

III. **Z** is a tri-atomic gas with a pungent smell. It has a V-shape. It is used in the manufacture of **Y**.

III. **Z:** SO_2

(06 marks x 3 = 18 marks)

(ii) Write the oxidation numbers and the electronic configurations of the two metals in **X**.

metal K oxidation number +I electronic configuration $1s^22s^22p^63s^23p^6$ or $1s^22s^22p^63s^23p^64s^1$

metal Cr oxidation number +VI electronic configuration $1s^22s^22p^63s^23p^6$ or $1s^22s^22p^63s^23p^63d^54s^1$

(02 marks x 6 = 12 marks)

(iii) I. Name the process by which **Y** is manufactured using **Z**.

Contact Process

.....

(02 marks)

II. When the gas formed on reaction of **Z** with $O_2(g)$ is dissolved in a concentrated solution of **Y**, the compound **P** is formed. When compound **P** reacts with water, **Y** is obtained again. Write the name and the chemical formula of compound **P**.

name Oleum / pyrosufluric acid / fuming sulfuric acid / disulfuric acid chemical formula ..... **(P)** $H_2S_2O_7$ .....

(04 marks x 2 = 08 marks)

(iv) Write the balanced chemical equation for the reaction taking place when **X**, **Y** and **Z** are reacted together.

$K_2Cr_2O_7(aq) + H_2SO_4(aq) + 3SO_2(g) \rightarrow K_2SO_4(aq) + Cr_2(SO_4)_3(aq) + H_2O(l)$

.....

(10 marks)

or

$Cr_2O_7^{2-}(aq) + 3SO_2(g) + 2H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3SO_4^{2-}(aq) + H_2O(l)$

(06 marks)

Physical states are not required.

2(a): 50 marks

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(*b*) A student is provided with eight bottles labelled **A**, **B**, **C**, **D**, **E**, **F**, **G** and **H** with aqueous solutions of $BaCl_2$, NaI, $Pb(NO_3)_2$, dil. HCl, $Al_2(SO_4)_3$, $AgNO_3$, conc. NH_4OH and dil. NH_4OH (not in order). Some useful observations for their identification on mixing two solutions at a time are given below.

	Solutions Mixed	Observation
I.	**A** + **C**	a yellow precipitate soluble in hot water
II.	**B** + **C**	a yellow precipitate insoluble in **H**
III.	**A** + **E**	a white precipitate soluble in hot water
IV.	**B** + **E**	a white precipitate soluble in **D**
V.	**E** + **F**	a white precipitate insoluble in **G**
VI.	**A** + **F**	a white precipitate insoluble in **G**
VII.	**D** + **G**	a colourless solution
VIII.	**H** + **G**	a colourless solution

(i) Identify **A** to **H**.

A $Pb(NO_3)_2$ **E** $BaCl_2$

B $AgNO_3$ **F** $Al_2(SO_4)_3$

C NaI **G** dil HCl

D dil NH_4OH **H** conc NH_4OH

(04 marks x 8 = 32 marks)

(ii) Give balanced chemical equations for the formation of precipitates, in each of the reactions I to VI. Use the symbol ↓ to indicate precipitates.

I. $Pb(NO_3)_2 + 2NaI \rightarrow PbI_2\downarrow + 2NaNO_3$

II. $AgNO_3 + NaI \rightarrow AgI\downarrow + NaNO_3$

III. $Pb(NO_3)_2 + BaCl_2 \rightarrow PbCl_2\downarrow + Ba(NO_3)_2$

IV. $2AgNO_3 + BaCl_2 \rightarrow 2AgCl\downarrow + Ba(NO_3)_2$

V. $3BaCl_2 + Al_2(SO_4)_3 \rightarrow 3BaSO_4\downarrow + 2AlCl_3$

VI. $3Pb(NO_3)_2 + Al_2(SO_4)_3 \rightarrow 3PbSO_4\downarrow + 2Al(NO_3)_3$

(03 marks x 6 = 18 marks)

Note: Deduct **01 mark** if precipitates are not indicated the using symbol ↓

2(b): 50 marks

3. (*a*) At temperature 25 °C, dissociation constants of $H_2CO_3(aq)$ acid are,

$K_1 = 4.5 \times 10^{-7}$ mol dm^{-3} and $K_2 = 4.7 \times 10^{-11}$ mol dm^{-3}

(i) Write the equilibrium reactions for the first and second dissociations of $H_2CO_3(aq)$.

i. **NOTE: PHYSICAL STATES ARE NECESSARY**

The symbol of equilibrium ⇌ also needed for all the reactions

1st dissociation: $H_2CO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$ ..... **(05)**

OR $H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$

2nd dissociation: $HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CO_3^{2-}(aq)$ ... **(05)**

OR $HCO_3^-(aq) \rightleftharpoons H^+(aq) + CO_3^{2-}(aq)$

(ii) Considering the first dissociation, calculate the concentrations of $H_3O^+(aq)$ and $HCO_3^-(aq)$ in 0.05 mol dm^{-3} $H_2CO_3(aq)$ solution at 25 °C.

ii. $K_1 = \dfrac{[H_3O^+(aq)][HCO_3^-(aq)]}{[H_2CO_3(aq)]}$ **(05)**

$H_2CO_3(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + HCO_3^-(aq)$

Initial con.	0.05	0	0	mol dm^{-3}
Change con.	$-x$	x	x	mol dm^{-3}
Equi. con.	$0.05 - x$	x	x	mol dm^{-3} **(05)**

$K_1 = 4.50 \times 10^{-7} = \dfrac{x \cdot x}{0.05 - x} \approx \dfrac{x^2}{0.05}$ **(05)**

$x^2 = 225 \times 10^{-10}$

$x = 1.5 \times 10^{-4}$ mol dm^{-3} **(04 + 01)**

$[H_3O^+(aq)] = [HCO_3^-(aq)] = 1.5 \times 10^{-4}$ mol dm^{-3} **(04 + 01)**

(iii) Considering the second dissociation, show that $[CO_3^{2-}(aq)]$ of the solution is approximately equal to K_2. State any assumption/s made.

iii. $K_2 = \dfrac{[H_3O^+(aq)][CO_3^{2-}(aq)]}{[HCO_3^-(aq)]}$ **(05)**

$HCO_3^-(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CO_3^{2-}(aq)$

Initial con.	1.5×10^{-4}	1.5×10^{-4}	0	mol dm^{-3}
Change con.	$-y$	$1.5 \times 10^{-4} + y$	y	mol dm^{-3}
Equi. con.	$1.5 \times 10^{-4} - y$	$1.5 \times 10^{-4} + y$	y	mol dm^{-3} **(05)***

$K_2 = 4.70 \times 10^{-11} = \dfrac{(1.5 \times 10^{-4} + y) \cdot y}{(1.5 \times 10^{-4} - y)} \approx y$ **(05)***

***If student writes in words that the amount dissociated from HCO_3^- is very small and amount accumulated in H_3O^+ is very small, therefore concentration change of the two species are negligible, award these 05 + 05 marks.**

$[CO_3^{2-}(aq)] \approx K_2$ **(05)**

Assumption: 2nd dissociation is much less compared to the 1st. **(05)**

3(a): 60 marks

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(*b*) You are provided with an aqueous solution containing 0.01 mol dm^{-3} $Al^{3+}(aq)$ ions and 0.01 mol dm^{-3} $Ag^{+}(aq)$ ions at temperature 25 °C. To 1.0 dm^3 of this solution, a concentrated solution of $PO_4^{3-}(aq)$ ions was added dropwise with continuous stirring.

At temperature 25 °C,

$K_{sp}(AlPO_4) = 1.3 \times 10^{-20}$ mol^2 dm^{-6} and $K_{sp}(Ag_3PO_4) = 8.1 \times 10^{-12}$ mol^4 dm^{-12}.

(i) Neglecting any volume change that may occur when the $PO_4^{3-}(aq)$ solution is added, state which metal ion (Al^{3+} or Ag^{+}) will precipitate first from the mixture. Give reasons for your answer based on a suitable calculation.

i. The precipitates formed are $AlPO_4$ ***and*** Ag_3PO_4

For $AlPO_4$

$$AlPO_4(s) \rightleftharpoons Al^{3+}(aq) + PO_4^{3-}(aq) \quad \textbf{(02)}$$

$$K_{sp} = [Al^{3+}(aq)][PO_4^{3-}(aq)] \quad \textbf{(02)}$$

$$1.30 \times 10^{-20} = 0.01 \times [PO_4^{3-}(aq)] \quad \textbf{(04)}$$

$[PO_4^{3-}(aq)] = 1.30 \times 10^{-18}$ mol dm^{-3} ; needed to precipitate $AlPO_4(s)$ **(03+01)**

For Ag_3PO_4

$$Ag_3PO_4(s) \rightleftharpoons 3Ag^{+}(aq) + PO_4^{3-}(aq) \quad \textbf{(02)}$$

$$K_{sp} = [Ag^{+}(aq)]^3[PO_4^{3-}(aq)] \quad \textbf{(02)}$$

$$8.10 \times 10^{-12} = (0.01)^3 \times [PO_4^{3-}(aq)] \quad \textbf{(04)}$$

$[PO_4^{3-}(aq)] = 8.10 \times 10^{-6}$ mol dm^{-3} ; needed to precipitate $Ag_3PO_4(s)$ **(03+01)**

$$[PO_4^{3-}(aq)]_{AlPO_4(s)} < [PO_4^{3-}(aq)]_{Ag_3PO_4(s)} \quad \textbf{(04)}$$

$AlPO_4(s)$ precipitates first. **(04)**

(ii) Calculate the concentration of the ion which precipitated first when the second ion starts to precipitate.

$[Al^{3+}(aq)]$ left in the solution after precipitation of $AlPO_4(s) = \frac{K_{sp(AlPO_4(s))}}{8.10 \times 10^{-6}}$ **(04)**

$$[Al^{3+}(aq)] \text{ left in the solution} = \frac{1.30 \times 10^{-20}}{8.10 \times 10^{-6}}$$

$$= 1.6 \times 10^{-15} \text{ mol dm}^{-3} \quad \textbf{(03+01)}$$

3(b): 40 marks

4. (a) ● The organic compound **A** having the molecular formula $C_5H_{10}O_3$ reacts with excess PCl_5 to give compound **B**, having a relative molecular mass of 155. Compound **A** liberates CO_2 with aqueous Na_2CO_3. (C = 12.0, H = 1.0, O = 16.0, Cl = 35.5)

(i) Write the functional groups present in compound **A**.

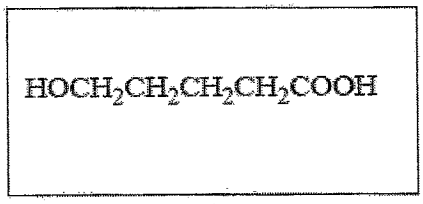
OH / hydroxy/ hydroxyl , COOH / carboxylic acid

Note: no marks if 'acid' is not written

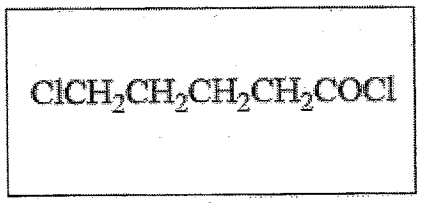
(05 marks x 2 = 10 marks)

● Compound **A** does not exhibit optical isomerism. Compound **A** reacts with pyridinium chlorochromate to give compound **C**. Compound **C** gives a silver mirror with ammoniacal $AgNO_3$. Compound **B** reacts with water to form compound **D**. Compound **D** reacts with alcoholic KOH to give the product **E** which contains a double bond.

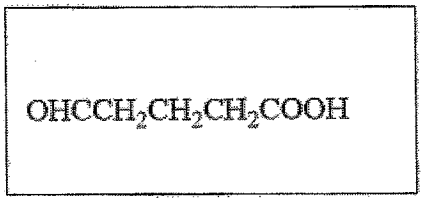
(ii) Draw the structures of **A**, **B**, **C**, **D** and **E** in the relevant boxes.



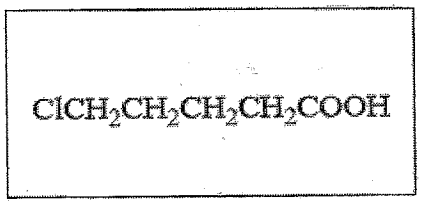
A



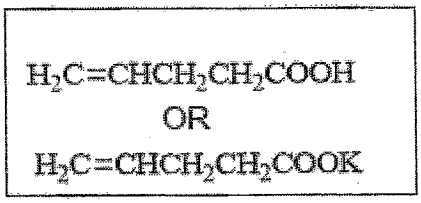
B



C



D



E

(07 marks x 5 = 35 marks)

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- Compound **F** is a structural isomer of **A**. Compound **F** reacts with excess PCl_5 to give compound **G** having a relative molecular mass of 155. Compound **F** does not liberate CO_2 with aqueous Na_2CO_3. Compound **F** exhibits optical isomerism. Compound **F** does not give a turbidity when treated with conc. HCl/anhydrous $ZnCl_2$. Compound **F** forms a coloured precipitate with 2,4-dinitrophenylhydrazine (2,4-DNP) and gives a silver mirror with ammoniacal $AgNO_3$.

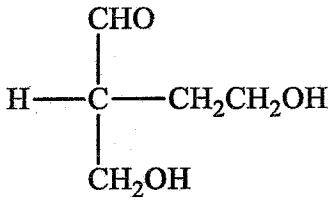
(iii) Write the functional groups present in **F**.

CHO / aldehyde , OH / hydroxy/ hydroxyl

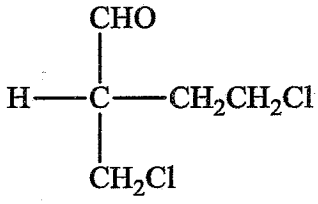
Note: If both are correct, award 09 marks. If only one is correct, award 05 marks only.

(05 marks + 04 marks = 09 marks)

(iv) Draw the structures of **F** and **G** in the relevant boxes.



(F)



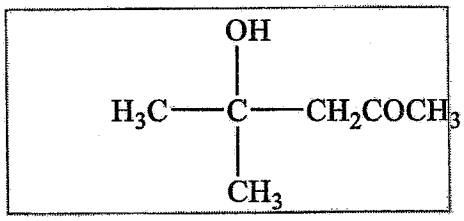
(G)

(07 marks x 2 = 14 marks)

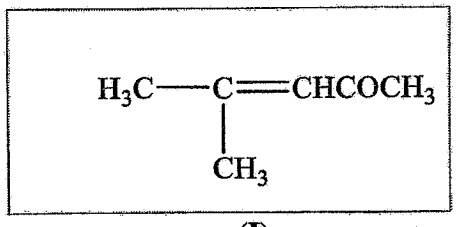
4(a): 68 marks

(*b*) (i) Draw the structure of the product **H** formed in the relevant box when acetone is reacted with dilute aqueous NaOH.

(ii) Draw the structure of the product **I** formed in the relevant box when **H** is heated with hydrohcloric acid.



(H)

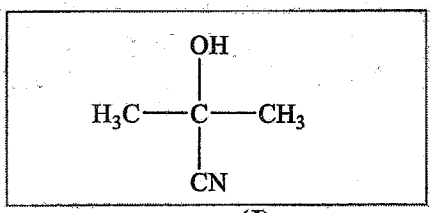


(I)

(06 marks x 2 = 12 marks)

4(b): 12 marks

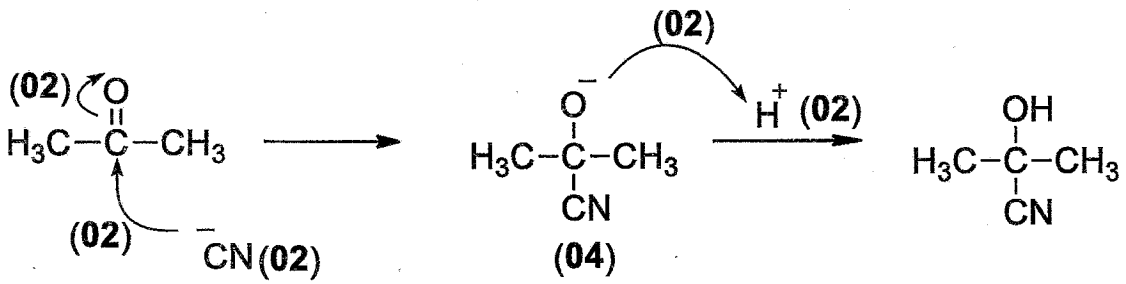
(c) (i) Draw the structure of the product **J** formed in the relevant box from the reaction between acetone and HCN.



(J)

(06 marks)

(ii) Write the mechanism of the above reaction.



Notes:

IF HCN is used instead of ^{-}CN deduct the **02** marks allocated for ^{-}CN.

(14 Marks)

4(c): 20 marks

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PART B — ESSAY

Answer two questions only. (Each question carries 150 marks.)

NOTE: PHYSICAL STATES ARE NECESSARY

5. (*a*) A mixture of NO(g) and O_2(g) in 2:1 molar ratio respectively, was introduced to a rigid close container of volume 10 dm^3 and allowed to react at temperature *T*. After a certain time, th system reached the equilibrium as given below, at temperature *T*.

$$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$$

At equilibrium, the following observations were noted.

- The pressure of the gaseous mixture was $32 \times 8.314 \times 10^3$ Pa.
- The total number of moles of the three gases was 0.64.
- The mass of O_2 was 6.4 g.

(i) Calculate the concentration of each gaseous species in mol dm^{-3} at equilibrium. (O = 16)

$$2NO(g) + O_2(g) \rightleftharpoons NO_2(g)$$

Moles of $O_2(g) = \frac{6.4\ g}{32\ g\ mol^{-1}} = 0.20$ **(02)**

Con. of $O_2(g) = \frac{0.20\ mol}{10\ dm^3} = 2.0 \times 10^{-2}$ mol dm^{-3} **(02+01)**

Moles of NO(g) = 0.40 (2:1 ratio given) **(02)**

Con. of $NO(g) = \frac{0.40\ mol}{10\ dm^3} = 4.0 \times 10^{-2}$ mol dm^{-3} **(02+01)**

Moles of $NO_2(g) = 0.64 - (0.40 + 0.20) = 0.04$ **(02)**

Con. of $NO_2(g) = \frac{0.04\ mol}{10\ dm^3} = 4.0 \times 10^{-3}$ mol dm^{-3} **(02+01)**

(ii) Calculate the equilibrium constant, K_c at temperature *T*.

$$K_C = \frac{[NO_2(g)]^2}{[O_2(g)][NO(g)]^2}$$ **(05)**

$$K_C = \frac{[4.0 \times 10^{-3}\ mol\ dm^{-3}]^2}{[2.0 \times 10^{-2}\ mol\ dm^{-3}][4.0 \times 10^{-2}\ mol\ dm^{-3}]^2}$$ **(05)**

$$\mathbf{K_C = 0.50\ mol^{-1}\ dm^3}$$ **(05)**

(iii) Calculate the value of temperature T (in K) under these conditions. State any assumption/s made.

Assuming ideal gas behavior **(03)**

$PV = nRT$ **(02)**

$T = \frac{PV}{nR}$

$T = \frac{32 \times 8.314 \times 10^3\ Pa \times 10 \times 10^{-3}\ m^3}{0.64\ \text{mol} \times 8.314\ \text{J}\ K^{-1}\ mol^{-1}}$ **(05)**

$T = 500$ K **(04+01)**

(iv) Calculate the equilibrium constant K_p for the reaction,

$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$

at the temperature determined in (iii) above.

Reaction of concern is

$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$ is the reverse reaction of the initial one **(05)**

If the equilibrium constant is K_C'

$K_C' = \frac{1}{K_C}$ **(05)**

$= \frac{1}{0.50} = 2\ \text{mol dm}^{-3}$ **(04+01)**

$K_P = K_C\ (RT)^{\Delta n} \qquad \Delta n = 1$ **(03+02)**

$K_P = 2\ (8.314\ \text{J}\ K^{-1}\ mol^{-1} \times 500\ \text{K})$

$K_P = 8.314 \times 10^3\ Pa$ **(04+01)**

5(a): 70 marks

(*b*) Consider the information given below at temperature 298 K.

$\Delta H_f^{\circ}(NO(g)) = 90\ \text{kJ mol}^{-1}$

$\frac{1}{2}N_2(g) + O_2(g) \rightarrow NO_2(g)\ \Delta H^{\circ} = 33\ \text{kJ mol}^{-1}$

$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)\ \Delta H^{\circ} = -102\ \text{kJ mol}^{-1}$

b. Information given:

$\Delta H_{f\ (NO(g))}^{\theta} = 90\ kJ\ mol^{-1}$ **(A)**

$\frac{1}{2}N_2(g) + O_2(g) \rightarrow NO_2(g)$: $\Delta H^{\theta} = 33\ kJ\ mol^{-1}$ **(B)**

$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)$: $\Delta H^{\theta} = -102\ kJ\ mol^{-1}$ **(C)**

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i. Reaction needed: $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

From (A), $\Delta H_{f\,(NO(g))}^{\theta}$: $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightarrow NO(g)$: $\Delta H^{\theta} = 90\ kJ\ mol^{-1}$ .. **(D)** **(05)**

(B) – (D) : $\frac{1}{2}O_2(g) + NO(g) \rightarrow NO_2(g)$ ..... **(E)** **(05)**

$\Delta H^{\theta} = (33 - 90) = -57\ kJ\ mol^{-1}$ **(05 +05)**

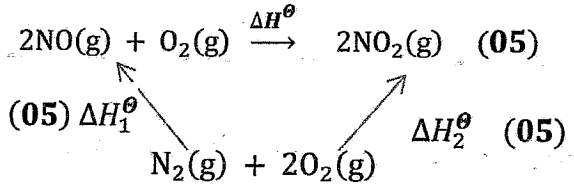
(E) × 2 : $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ $\Delta H^{\theta} = -114\ kJ\ mol^{-1}$ **(04+01)**

OR: **(B) × 2 – (D) × 2** : $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ **(05+05)**

$\Delta H^{\theta} = (66 - 180)\ kJ\ mol^{-1}$ **(05+05)**

$\Delta H^{\theta} = -114\ kJ\ mol^{-1}$ **(04+01)**

OR



From Hess's law: $\Delta H^{\theta} = \Delta H_2^{\theta} - \Delta H_1^{\theta} = (2 \times 33 - 2 \times 90) kJ\ mol^{-1}$ **(05)**

$= -114\ kJ\ mol^{-1}$ **(04 + 01)**

OR other acceptable method(s)

ii. Formation reaction of $2N_2O_5(g)$:

$N_2(g) + \frac{5}{2}O_2(g) \rightarrow N_2O_5(g)$: $\Delta H_f^{\theta} = ?$ **(05)**

Information: $\frac{1}{2}N_2(g) + \frac{1}{2}O_2(g) \rightarrow NO(g)$: $\Delta H^{\theta} = 90\ kJ\ mol^{-1}$ ..... **(D)**

(D) × 2 : $N_2(g) + O_2(g) \rightarrow 2NO(g)$: $\Delta H^{\theta} = 180\ kJ\ mol^{-1}$ ..... **(F)** **(05+05)**

$2NO(g) + O_2(g) \rightarrow NO_2(g)$ $\Delta H^{\theta} = -114\ kJ\ mol^{-1}$ .... **(G)**

$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)$: $\Delta H^{\theta} = -102\ kJ\ mol^{-1}$ **(C)**

(C) ÷ 2 : $2NO_2(g) + \frac{1}{2}O_2(g) \rightarrow N_2O_5(g)$: $\Delta H^{\theta} = -51\ kJ\ mol^{-1}$ ..... **(H)** **(05)**

(F) + (G) + (H): $N_2(g) + \frac{5}{2}O_2(g) \rightarrow N_2O_5(g)$

$\Delta H_{f\,(N_2O_5(g))}^{\theta} = [180 + (-114) + (-51)]\ kJ\ mol^{-1} =$ **15 kJ mol^{-1}** **(05+04+01)**

OR

$$4NO_2(g) + O_2(g) \xrightarrow{-102} 2N_2O_5(g) \quad \textbf{(05)}$$

$$\textbf{(05)}\ 4 \times 33 \qquad 2\Delta H^{\theta}_{f(N_2O_5(g))}\ \textbf{(05)}$$

$$2N_2(g) + 5O_2(g)$$

From Hess's law: $2\Delta H^{\theta}_{f(N_2O_5(g))} = (4 \times 33 - 102) kJ\ mol^{-1}$ **(05 + 05)**

$$\Delta H^{\theta}_{f(N_2O_5(g))} = \mathbf{15}\ kJ\ mol^{-1}\ \textbf{(04 + 01)}$$

OR other acceptable method(s)

(iii) Using the results obtained in (ii) above, predict the following.

I. the sign of $\Delta S^{\circ}_{f}(N_2O_5(g))$

$$N_2(g) + \frac{5}{2}O_2(g) \rightarrow N_2O_5(g)$$

3.5 moles → 1 mole : number of moles decreased **(05)**

$\Delta H^{\theta}_{s\,(N_2O_5(g))}$ is negative (< 0) **(05)**

II. spontaneity of the reaction for the formation of $N_2O_5(g)$ from $N_2(g)$ and $O_2(g)$

$\Delta G^{\theta} = \Delta H^{\theta} - T\Delta S^{\theta}$ **(05)**

= +ve – (+ve)(-ve) = +ve **(05)**

The reaction is non-spontaneous at 298 K **(05)**

5(b): 80 marks

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NOTE: PHYSICAL STATES ARE NECESSARY

6. (*a*) According to the kinetic molecular theory of gases, for an ideal gas at temperature T, $PV = \frac{1}{3}mN\overline{C^2}$. Here P is the pressure of the gas, V is the volume of the gas, m is the mass of a gas molecule, N is the number of molecules of gas and $\overline{C^2}$ is the mean square speed of the gas.

(i) Show that $\overline{C^2} = \frac{3RT}{M}$ for an ideal gas. M is the molar mass of the gas.

$PV = \frac{1}{3}mN\,\overline{C^2} = \frac{1}{3}m(nN_A)\overline{C^2}$: n – moles, N_A – Avogadro no **(04)**

$= \frac{1}{3}nM\overline{C^2}$: M-molar mass **(04)**

$PV = nRT = \frac{1}{3}nM\overline{C^2}$ **(04)**

$\overline{C^2} = \frac{3RT}{M}$ **(04)**

(ii) **A** and **B** are two ideal gases with molar masses M_A and M_B respectively. Show that the mean square speed of gas **B** $\left(\overline{C_B^2}\right)$ at temperature $T = 300\frac{M_B}{M_A}$, is equal to the mean square speed of gas **A** $\left(\overline{C_A^2}\right)$ at $T = 300$. (Temperatures are given in kelvin)

For gas A: $\overline{C_A^2} = \frac{3RT}{M_A} = \frac{3R \times 300}{M_A}$ **(1)**

For gas B: $\overline{C_B^2} = \frac{3RT}{M_B} = \frac{3R \times 300(\frac{M_B}{M_A})}{M_B} = \frac{3R \times 300}{M_A}$ **(2)** **(04+04)**

(1) = (2): $\overline{C_A^2} = \overline{C_B^2}$ **(04)**

(iii) Derive an expression for the ratio between the molar kinetic energies of the two gases **A** and **B** at any given temperature T.

For gas A: $(KE)_A = \frac{3RT_A}{2}$ **(04)**

For gas B: $(KE)_B = \frac{3RT_B}{2}$ **(04)**

As $T_A = T_B$: $(KE)_A = (KE)_B$ **(04)**

OR- Kinetic energy of a gas depends only on the absolute temperature

$(KE)_A = (KE)_B$ **(12)**

6(a): 40 marks

(*b*) (i) Define the term 'an elementary reaction'.

Elementary Reaction: A reaction occurs through a single step (without involving an intermediate). **(05)***

(ii) Define the term 'molecularity' of a reaction.

Molecularity: Total number of reacting molecules participating in an elementary reaction or in the rate determining step of a reaction. **(05)**

(iii) For an elementary reaction what is the relationship between 'reaction order' and 'molecularity'?

Elementary reaction : Reaction order = Molecularity **(05)***

(iv) The following table gives the variation of the concentration of the reactant in a reaction with time.

Time (minutes)	0	10	20	30	40
Reactant concentration (mol dm^{-3})	1.6	0.8	0.4	0.2	0.1

Reaction order = Molecularity

In 10 minutes time the concentration decreased to half of the initial concentration **(05)**

This is consistent with all the 10 min time intervals **(05)**

I) Reaction is first order **(05)**

II) Half-life ($t_{1/2}$) of the reaction is 10 min **(05)**

*** Note: If student has attempted Q 6 (b), award marks for parts i and iii**

(v) Consider the information given below for two first order reactions ① and ② at a given temperature.

	Reaction	Reaction rate/ mol dm^{-3} s^{-1}	Rate constant/s^{-1}	Half-life/s
①:	$A \rightarrow P_1$	r_A	k_A	$(t_{1/2})_A$
②:	$B \rightarrow P_2$	r_B	k_B	$(t_{1/2})_B$

(P_1, P_2 = Products)

A first order reaction with rate constant, k has a half-life, $t_{1/2} = \frac{0.693}{k}$.

If $r_B = 3r_A$ when [B] = 2[A], show that $2(t_{1/2})_A = 3(t_{1/2})_B$.

As the reactions are first order,

$r_A = k_A[A]$: $k_A = \frac{r_A}{[A]}$ **(03+02)**

$r_B = k_B[B]$: $k_B = \frac{r_B}{[B]}$ **(03+02)**

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Given that: [B] = 2[A], $r_B = 3r_A$ *and* $t_{1/2} = \frac{0.693}{k}$

We can write; $k_A = \frac{r_A}{[A]}$ .....(I)

$k_B = \frac{3r_A}{2[A]}$ .....(II) **(05)**

(I)÷(II) gives $\frac{k_A}{k_B} = \frac{2}{3}$ **(05+05)**

$\frac{(t_{1/2})_A}{(t_{1/2})_B} = \frac{0.693/k_A}{0.693/k_B} = \frac{k_B}{k_A} = \frac{3}{2}$ **(05+05)**

$$\mathbf{2(t_{1/2})_A = 3(t_{1/2})_B}$$ **(05)**

6(b): 75 marks

(c) At temperature 25 °C, 50.0 cm^3 of 0.30 g dm^{-3} iodine aqueous solution was shaken well with 10.0 cm^3 of CCl_4. When the system reached equilibrium the concentration of iodine in the water layer was found to be 0.02 g dm^{-3}.

(i) Calculate the concentration of iodine in the CCl_4 layer at equilibrium.

Initial I_2 mass in water layer = 0.30 g dm^{-3} × 50 × 10^{-3} dm^3 = 0.015 g **(03)**

Equilibrium I_2 mass in water layer = 0.02 g dm^{-3} × 50 × 10^{-3} dm^3 = 0.001 g **(03)**

Equilibrium $[I_2]_{H_2O} = \frac{(0.001\ g)}{50 \times 10^{-3} dm^3} = 0.02\ \text{g}\ dm^{-3}$

Equilibrium I_2 mass in CCl_4 layer $= (0.015 - 0.001)g = 0.014\ g$ **(03)**

Equilibrium $[I_2]_{CCl_4} = \frac{(0.014\ g)}{10 \times 10^{-3} dm^3} = \mathbf{1.4\ g\ dm^{-3}}$ **(03+03+01)**

(ii) At temperature 25 °C, calculate the partition co-efficient of I_2 between CCl_4 and water.

$K_D = \frac{[I_2]CCl_4}{[I_2]_{H_2O}} = \frac{1.4\ \text{g}\ dm^{-3}}{0.02\ \text{g}\ dm^{-3}} = \mathbf{70}$ **(03+03)**

(iii) If the above experiment was done at 25 °C with 20.0 cm^3 of CCl_4 instead 10.0 cm^3, calculate the concentration of iodine in the water layer at equilibrium.

At this stage take I_2 mass in H_2O layer as x

$K_D = 70 = \frac{[I_2]CCl_4}{[I_2]_{H_2O}} = \frac{(0.015-x)/20}{x/50}$ **(05)**

$x = 0.0005\ g$

(04) $[I_2]_{H_2O} = \frac{(0.0005\ g)}{50 \times 10^{-3} dm^3} = \mathbf{0.01\ g\ dm^{-3}}$ **(03+01)**

Alternate answer:

At this stage take I_2 mass in CCl_4 layer as x

$$K_D = 70 = \frac{[I_2]_{CCl_4}}{[I_2]_{H_2O}} = \frac{x/20}{(0.015-x)/50}$$ **(05)**

$x = 0.0145\ g$ **(02)**

I_2 mass in water layer = 0.0150 – 0.0145 = 0.0005 g **(02)**

$$[I_2]_{H_2O} = \frac{(0.0005\ g)}{50 \times 10^{-3} dm^3} = \mathbf{0.01\ g\ dm^{-3}}$$ **(03+01)**

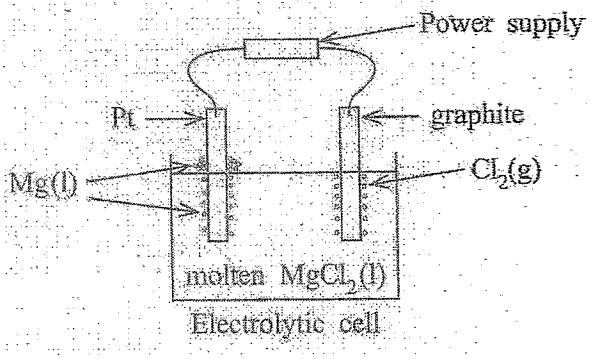
6(c): 35 marks

NOTE: PHYSICAL STATES ARE NECESSARY

7.(a) Mg metal can be extracted by the electrolysis of molten $MgCl_2(l)$ using inert electrodes (examples :Pt, graphite). A simple setup for this is shown in the diagram.

$E^\circ_{Mg^{2+}(l)/Mg(s)} = -2.37$ V

$E^\circ_{H_2O(l)/H_2(g)} = -0.63$ V



(i) Identify the anode and the cathode. Write the half reaction taking place at each electrode.

	Electrode	Half-reaction	
Anode :	C(graphite, s)|Cl_2(g)|$MgCl_2(l)$	$2Cl^-(l) \rightarrow Cl_2(g) + 2e$	**(05+05)**
Cathode :	Pt(s)|Mg(s)|$MgCl_2(l)$	$Mg^{2+}(l) + 2e \rightarrow Mg(s)$ or $Mg(l)$	**(05+05)**
OR		$MgCl_2(l) \rightarrow Mg(s)/(l) + Cl_2(g)$	**(05)**

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(ii) Write the overall cell reaction.

Overall cell reaction:

$$Mg^{2+}(l) + 2Cl^{-}(l) \rightarrow Mg(s)/(l) + Cl_2(g)$$

OR $$MgCl_2(l) \rightarrow Mg(s)/(l) + Cl_2(g)$$ **(05)**

(iii) State the direction of electron flow through the external circuit as the cell operates.

from C(graphite, s)|Cl_2(g) anode to Pt(s)|Mg(s) cathode **(05)**

(iv) Explain the following:

I. Molten $MgCl_2$(l) is used instead of $MgCl_2$(s) in this extraction process.

Solid or crystalline $MgCl_2(s)$ has no movable ions. Molten state has ionic $Mg^{2+}(l)$ and $Cl^{-}(l)$. **(05)**

II. A solution of $MgCl_2$(aq) cannot be used in this extraction process.

Water molecules would reduce to H_2(g) instead of $Mg^{2+}(l)$ **(05)**

(v) If a 5.37 A current is passed through this cell for one hour and the Cl_2(g) formed is collected at temperature 300 K and pressure 1 atm (~ 1.0×10^5 Pa), calculate the volume

$Q = It$ **(05)**

$= 5.37\ A \times (60 \times 60) = 19300\ C$ **(04+01)**

Moles of electrons = 19300/96500 = 0.2 mol **(05)**

2 mol of electrons ≡ 1 mol of Cl_2(g)

Moles of Cl_2(g) produced = 0.10 mol **(05)**

Assuming an ideal gas behavior: PV = nRT **(05)**

$$V = \frac{nRT}{P} = \frac{0.10\ mol \times 8.314\ J\ K^{-1}\ mol^{-1} \times 300\ K}{1 \times 10^5\ Pa}$$ **(05)**

$$= \mathbf{249.4 \times 10^{-5}\ m^3\ or\ 2.49\ dm^3}$$ **(04+01)**

7(a): 75 marks

(*b*) (i) **P**, **Q**, **R**, **S** and **T** are coordination compounds of Co(III). They have an octahedral geometry. Give the structural formulae or draw the structures of these coordination compounds, selecting the appropriate species from the list given below.

Co^{3+}, K^+, NH_3, SO_4^{2-}, NO_2^-, Cl^-

Note : In the above coordination compounds NO_2^- behaves as a monodentate ligand when attached to the metal ion.

P – Only neutral ligands are coordinated to the metal ion. On reaction of an aqueous solution of **P** with dil. HCl, reddish-brown fumes are evolved. **P** gives four ions in aqueous solution.

Q – Two types of ligands are coordinated to the metal ion. They are neutral ligands and mono-atomic anionic ligands. A white precipitate insoluble in dilute acid is formed on addition of $BaCl_2(aq)$ to an aqueous solution of **Q**. **Q** gives two ions in aqueous solution.

R – Two types of ligands are coordinated to the metal ion. They are neutral ligands and multi-atomic anionic ligands. **R** shows geometric isomerism. On reaction of an aqueous solution of **R** with $AgNO_3(aq)$, a white precipitate is formed. This precipitate is soluble in dil. NH_4OH. **R** gives two ions in aqueous solution.

S – It is a non-ionic compound. An equal number of neutral ligands and multi-atomic anionic ligands are coordinated to the metal ion.

T – Only mono-atomic anionic ligands are coordinated to the metal ion. **T** gives four ions in aqueous solution.

P : $[Co(NH_3)_6](NO_2)_3$ **(10)**

Q: $[Co(NH_3)_5Cl]SO_4$ **(10)**

R: $[Co(NH_3)_4(NO_2)_2]Cl$ **(10) ***

S: $[Co(NH_3)_3(NO_2)_3]$ **(10)**

T: $K_3[CoCl_6]$ **(10)**

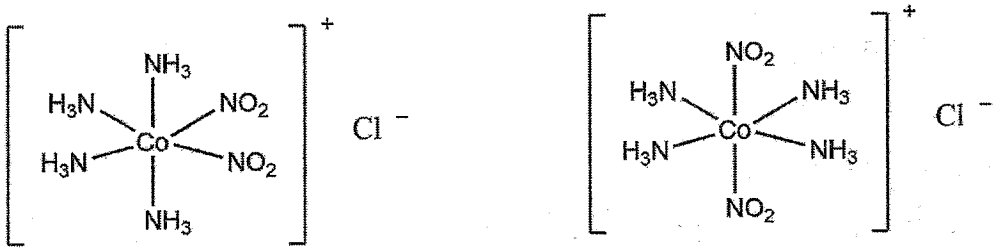
(7 (b) (i) 50 marks)

(ii) I. Write the IUPAC name of **T**.

potassium hexachloridocobaltate(III) **(10)**

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II. Draw the structures of the geometric isomers of **R**.



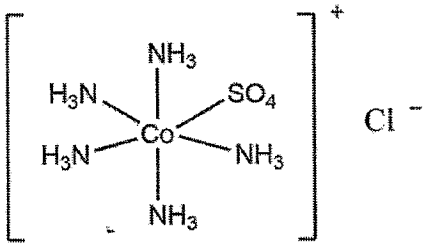
(05 marks)*

(If both are correct 05 marks. If only one is correct 03 marks)

***Note: Although geometric isomerism is commonly taught in A/Levels, students may not be familiar with them in coordinate complexes. Therefore, as a concession, marks will be awarded in the following manner for 7 b (i) R and (ii) II.**

(i) R: $[Co(NH_3)_5(SO_4)]Cl$ **(06)**

(ii) II.



(03)

(iii) **X** is a coordination compound of Co(III) with an octahedral geometry. The ligands H_2O and CO_3^{2-} are coordinated to the metal ion. On treatment of an aqueous solution of **X** with $AgNO_3(aq)$ a pale yellow precipitate, soluble in conc. NH_4OH is formed. **X** gives two ions in aqueous solution. Give the structural formula or draw the structure of **X**.

Note : CO_3^{2-} coordinates to the metal ion through two oxygen atoms.

$[Co(H_2O)_4(CO_3)]Br$ **(10)**

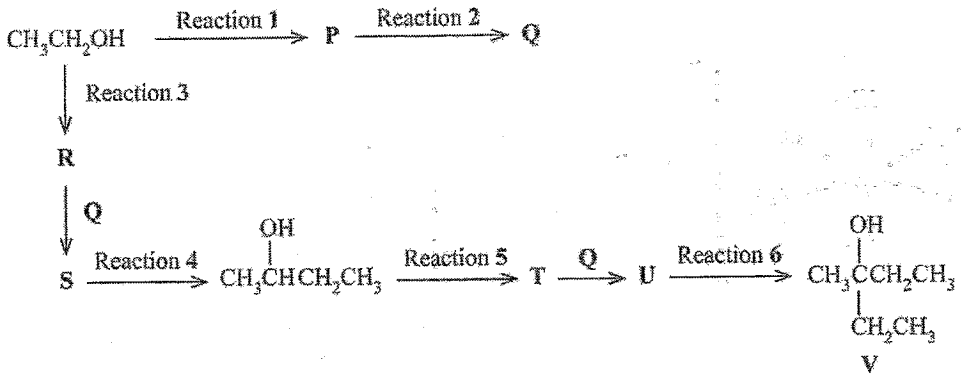
7(b): 75 marks

PART C — ESSAY

Answer two questions only. (Each question carries 150 marks.)

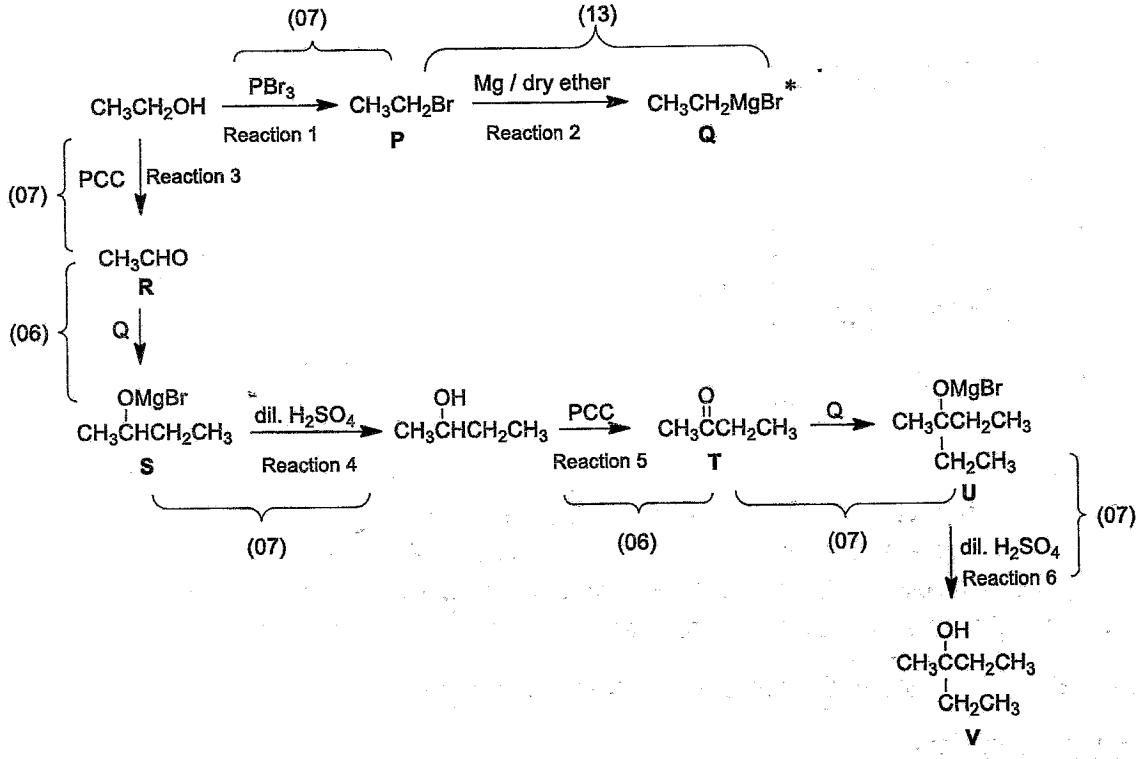
8. (*a*) Given below is a reaction scheme for the preparation of compound **V** using ethanol as the only organic starting material.

8. (a)



Complete the above reaction scheme by drawing the structures of compounds **P**, **Q**, **R**, **S**, **T** and **U** and writing the appropriate reagents for reactions **1** - **6** selected **only** from those given in the list below.

Reagents:
Dilute H_2SO_4, Mg/dry ether, PBr_3, Pyridinium chlorochromate (PCC)



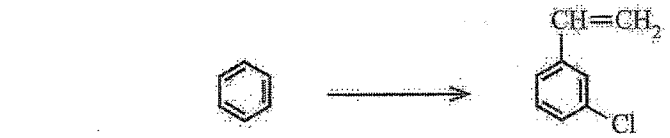
IF the structure of **Q is not given or wrong but given **correctly** as a reagent in an appropriate place, award **06** marks once.*

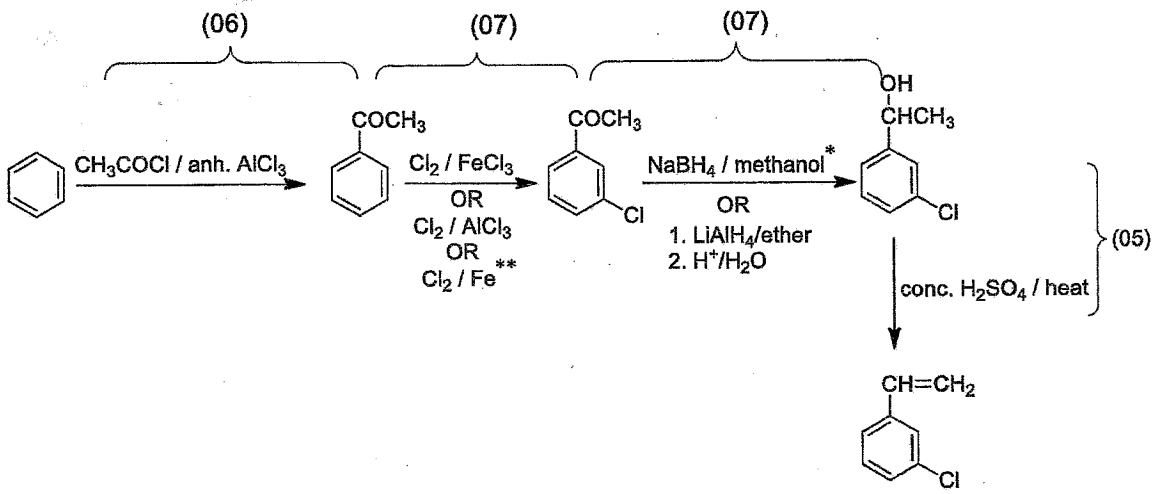
***NOTE:** If the structure of **Q** is wrong or not given, mark the rest of the answer as given in the marking scheme.*

8(a): 60 marks

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(*b*) (i) Show how you would carry out the following conversion in not more than **four (04)** steps.





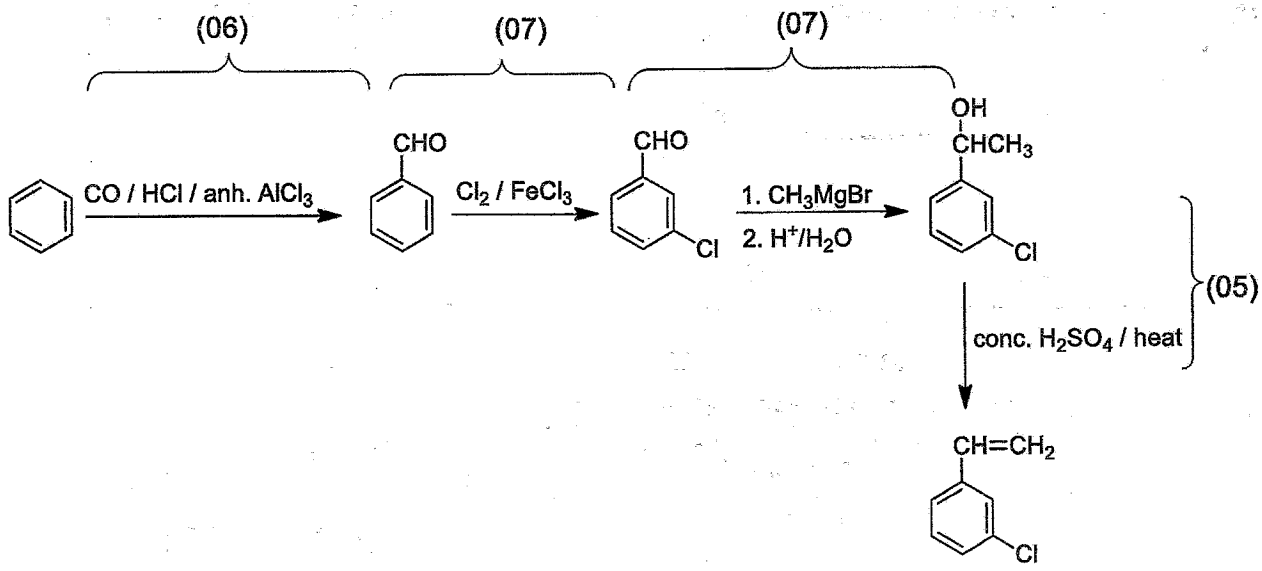
Methanol is necessary for the reaction. If methanol is not written deduct **01 mark.*

***This is obsolete and not given in the syllabus or in the teacher guide. But it seems that it has been continuously taught in certain schools. Hence in fairness to the students it is included in the marking scheme.*

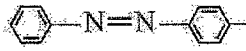
(25 marks)

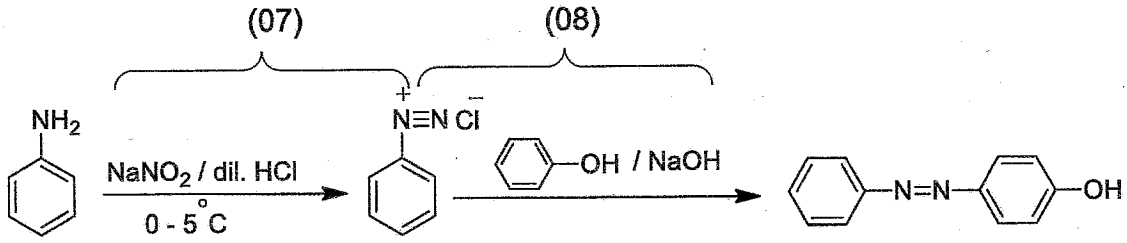
Alternative answer:

Although the conversion of benzene to benzaldehyde is not given in the syllabus or in the teacher guide, it seems that it has been continuously taught in certain schools. Hence in fairness to the students the following alternative answer is included in the marking scheme.



(25 marks)

(ii) Propose a method to prepare  from aniline in not more than two **(02)** steps.



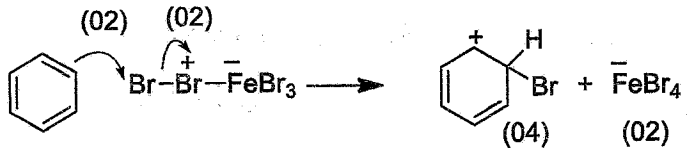
(15 marks)

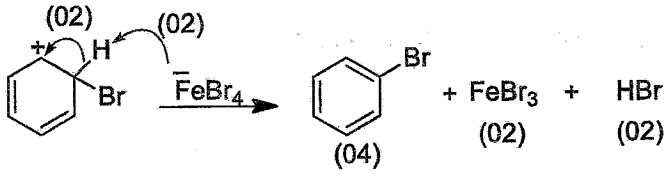
8(b): 40 marks

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(c) (i) Write the product and the mechanism of the reaction that takes place between benzene and bromine in the presence of anhydrous $FeBr_3$.

$Br_2 + FeBr_3 \longrightarrow Br-\overset{+}{B}r-\overset{-}{F}eBr_3$ or $Br^{+}\ \overset{-}{F}eBr_4$ or $Br-\overset{+}{B}r\ \overset{-}{F}eBr_3$ or $\overset{\delta+}{Br}-\overset{\delta-}{Br}\ FeBr_3$ (04)





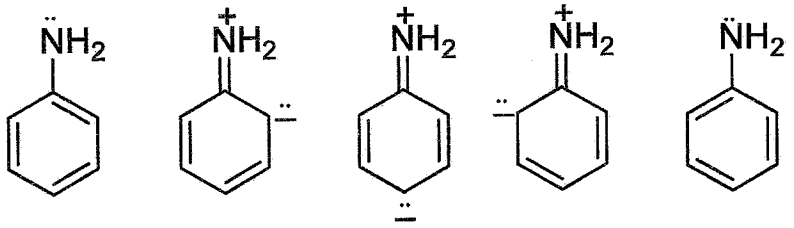
(26 marks)

(ii) Draw the resonance structures of benzene and aniline.

Resonance structures of benzene



Resonance structures of aniline



Note: **IF** the lone pair is not written, deduct **02** marks once.

(02 marks x 7 = 14 marks)

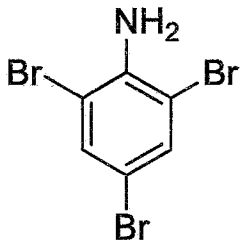
(iii) Considering the above resonance structures, explain why the benzene nucleus in aniline is more reactive towards electrophilic substitution reactions than benzene itself.

The benzene ring in aniline is electron rich compared to benzene (03 marks) due to the delocalization of the lone pair of electrons on the nitrogen atom, into the benzene ring in aniline (03 marks).

Therefore, benzene ring in aniline is more reactive towards electrophiles, than benzene.

(06 Marks)

(iv) Draw the structure of the product formed when aniline reacts with bromine.



(04 marks)

8(c): 50 marks

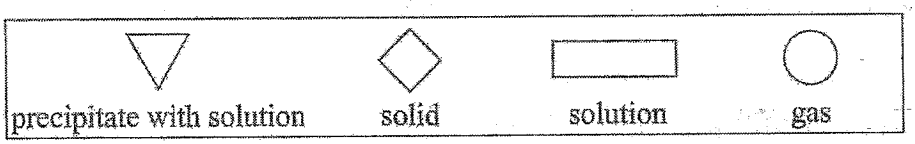
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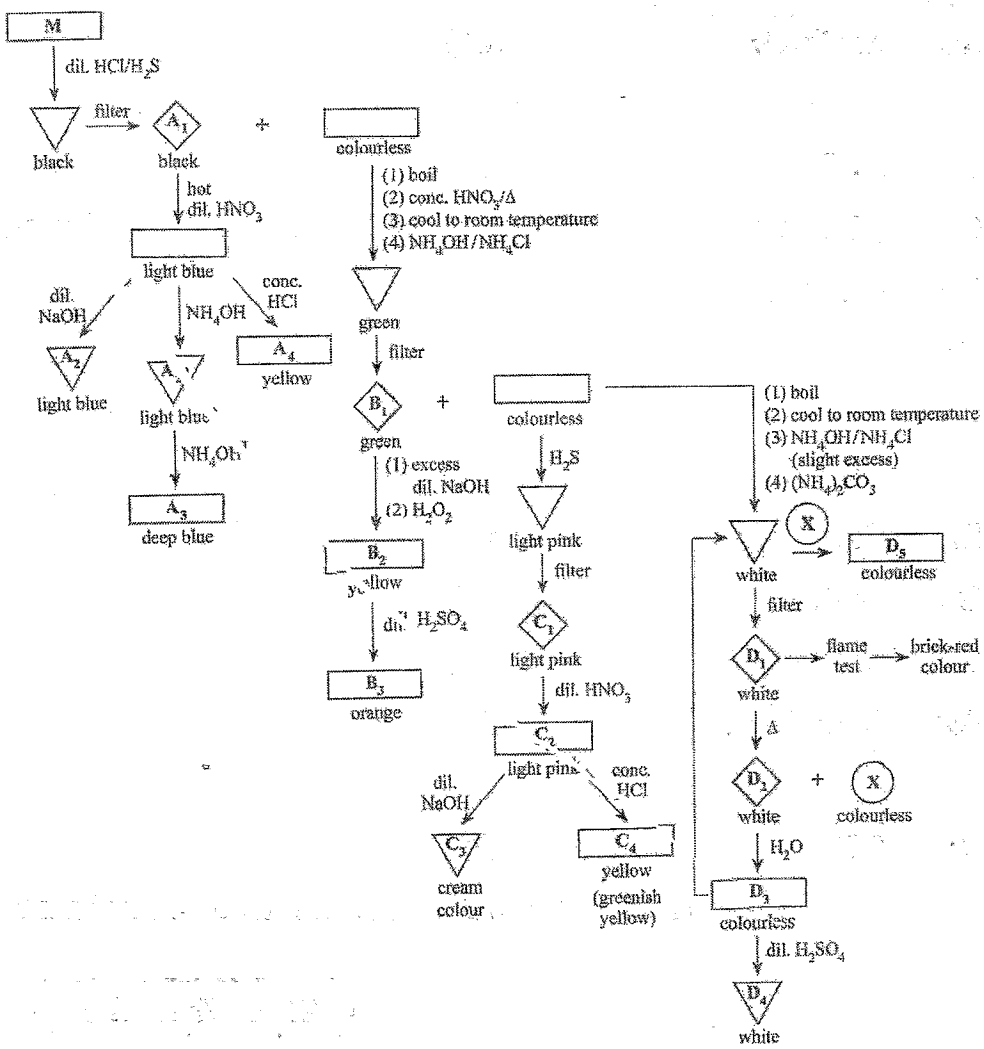
9. (*a*) The following question is based on the qualitative analysis of cations.

An aqueous solution **M** contains one cation of each of the metals **A**, **B**, **C** and **D**.

M is analysed according to the scheme given below.

The symbols given in the box are used to represent precipitate with solution, solids, solutions and gases.





A_1–A_4, B_1–B_3, C_1–C_4 and D_1–D_5 are compounds/species of the four cations of metals **A**, **B**, **C** and **D**. **X** is a gas.

Identify A_1, A_2, A_3, A_4, B_1, B_2, B_3, C_1, C_2, C_3, C_4, D_1, D_2, D_3, D_4, D_5, and **X**.

(**Note** : Write **only** chemical formulae. Chemical equations and reasons are **not** required.)

A_1:	CuS	**(06)**
A_2:	$Cu(OH)_2$	**(04)**
A_3:	$[Cu(NH_3)_4]^{2+}$	**(04)**
A_4:	$[CuCl_4]^{2-}$	**(04)**
B_1:	$Cr(OH)_3$	**(06)**
B_2:	Na_2CrO_4	**(04)**
B_3:	$Na_2Cr_2O_7$	**(04)**
C_1:	MnS	**(06)**
C_2:	$[Mn(H_2O)_6]^{2+}$	**(04) Note:** If $Mn^{2+}(aq)$ only **02 marks**
C_3:	$Mn(OH)_2$	**(04)**
C_4:	$[MnCl_4]^{2-}$	**(04)**
D_1:	$CaCO_3$	**(06)**
D_2:	CaO	**(04)**
D_3:	$Ca(OH)_2$	**(04)**
D_4:	$CaSO_4$	**(04)**
D_5:	$Ca(HCO_3)_2$	**(04)**
X:	CO_2	**(03)**

(06 marks x 4 + 04 marks x 12 + 03 marks x 1)

9(a): 75 marks

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(*b*) The main compound present in iron pyrite is FeS_2. A 1.50 g sample of iron pyrite was oxidized under laboratory conditions and all the sulphur in FeS_2 was converted to SO_4^{2-}. The resultant SO_4^{2-} was precipitated as $BaSO_4$. The dry weight of $BaSO_4$ obtained was 4.66 g.

(i) Calculate the weight percentage of FeS_2 present in iron pyrite.

H_2SO_4 that was produced in this reaction after 120 hours was quantitatively separated and precipitated as $BaSO_4$. The dry weight of $BaSO_4$ obtained was 31.13 g.

FeS_2 in 20.0 g of iron pyrite was subjected to oxidation by soil microbes under natural conditions for 120 hours.

This oxidation reaction is represented in the following equation.

$$4FeS_2(s) + 15O_2(g) + 14H_2O(l) \xrightarrow{\text{soil microbes}} 4Fe(OH)_3(s) + 8H_2SO_4(aq)$$

$BaSO_4$ - Molar mass = 137 + 32 + 64 = 233 g mol^{-1} **(03)**

FeS_2 - Molar mass = 56 + 64 = 120 g mol^{-1} **(03)**

Moles of $BaSO_4 = \frac{4.66}{233}$ = 0.02 mol **(03)**

1 mole of FeS_2 gives 2 moles $BaSO_4$ **(03)**

$\therefore$ moles of $FeS_2 = \frac{0.02}{2}$ = 0.01 mol **(03)**

$\therefore$ weight of FeS_2 = 0.01 x 120 = 1.20 g **(03)**

$\therefore$ weight percentage of FeS_2 in iron pyrite $= \frac{1.20}{1.50} \times 100$ **(03)**

$= 80\%$ **(04)**

(9b(i): 25 marks)

(ii) Calculate the percentage conversion of FeS_2 in iron pyrite to SO_4^{2-} after 120 hours by soil microbes.

Note : Percentage conversion $= \frac{\text{Experimentally obtained mass using soil microbes}}{\text{Theoretical mass}} \times 100$

4 moles $FeS_2 \rightarrow$ 8 moles $H_2SO_4 \rightarrow$ 8 moles $BaSO_4$

Amount of FeS_2 in 20.0 g of iron pyrite $= 20.0 \times \frac{80}{100}$ **(03)**

$= 16.0$ g **(03)**

Calculation of theoretical mass of $BaSO_4$

According to the equation

4 moles FeS_2 gives 8 moles of $BaSO_4$ **(03)**

$\therefore$ 4 x 120 g FeS_2 gives 8 x 233 g of $BaSO_4$ **(03)**

$\therefore$ 16 g FeS_2 gives $\frac{8 \times 233}{4 \times 120} \times 16$ g of BaSO4 **(03)**

Theoretical mass of $BaSO_4$ = 62.13 g **(03)**

Experimental mass of $BaSO_4$ = 31.13 g

% conversion $= \frac{31.13}{62.13} \times 100$ **(03)**

$= 50.1\%$ **(04)**

(9b(ii): 25 marks)

ALTERNATE ANS 1

4 moles $FeS_2 \rightarrow$ 8 moles $H_2SO_4 \rightarrow$ 8 moles $BaSO_4$

Mass of FeS_2 in 20.0 g of iron pyrite $= 20.0 \times \frac{80}{100}$ **(03)**

$= 16.0$ g **(03)**

Moles of $BaSO_4$ $= \frac{31.13}{233}$ mol

Moles of H_2SO_4 $= \frac{31.13}{233}$ mol **(03)**

Moles of FeS_2 reacted $= \frac{1}{2} \times \frac{31.13}{233}$ mol **(03)**

Experimental mass of FeS_2 reacted $= \frac{1}{2} \times \frac{31.13}{233}$ mol x 120 g mol^{-1} **(03)**

$= 8.016$ g **(03)**

% conversion $= \frac{8.016g}{16g} \times 100\%$ **(03)**

$= 50.1\%$ **(04)**

(9b(ii): 25 marks)

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ALTERNATE ANS 2

4 moles $FeS_2 \rightarrow$ 8 moles $H_2SO_4 \rightarrow$ 8 moles $BaSO_4$

Mass of FeS_2 in 20.0 g of iron pyrite $= 20.0\ g \times \frac{80}{100}$ **(03)**

$= 16.0$ g **(03)**

Moles of FeS_2 $= \frac{16.0}{120}$ mol **(03)**

Theoretical mass of $BaSO_4$ $= \frac{16}{120}$ x 2 mol x 233 g mol^{-1} **(03)**

$= 62.13$ g **(03)**

Experimental mass of $BaSO_4$ reacted $= 31.13$ g **(03)**

% conversion $= \frac{31.13g}{62.13g}$ x 100% **(03)**

$= 50.1\%$ **(04)**

(9b(ii): 25 marks)

ALTERNATE ANS 3

4 moles $FeS_2 \rightarrow$ 8 moles $H_2SO_4 \rightarrow$ 8 moles $BaSO_4$

Mass of FeS_2 in 20.0 g of iron pyrite $= 20.0\ g \times \frac{80}{100}$ **(03)**

$= 16.0$ g **(03)**

Moles of FeS_2 $= \frac{16.0}{120}$ mol **(03)**

Theoretical moles of H_2SO_4 $= \frac{16}{120}$ x 2 mol

Theoretical mass of H_2SO_4 $= \frac{16}{120}$ x 2 mol x 98 g mol^{-1}

$= 26.13$ g **(03)**

Moles of BaSO4 $= \frac{31.13}{233}$ mol

Moles of H_2SO_4 $= \frac{31.13}{233}$ mol

Experimental mass of H_2SO_4 $= \frac{31.13}{233}$ mol x 98 g mol^{-1} **(03)**

= 13.09 g (03)

% conversion $= \frac{13.09g}{26.13g} \times 100\%$ (03)

= 50.1% (04)

(9b(ii): 25 marks)

(iii) Calculate the amount of iron pyrite required to produce 8 kg of H_2SO_4 by soil microbes when the conversion percentage of FeS_2 in iron pyrite to SO_4^{2-} is 100%.
(Relative atomic mass : O = 16, S = 32, Fe = 56, Ba = 137)

H_2SO_4 = 8 kg = 8000 g (03)

Molar mass H_2SO_4 = 98 (03)

Moles $H_2SO_4 = \frac{8000}{98}$ (03)

If the conversion is 100%

$\therefore$ moles of FeS_2 required $= \frac{8000}{98} \times \frac{1}{2}$ (03)

= 40.8 moles (03)

= 40.8 x 120 g

= 4896 g (03)

$\therefore$ amount of iron pyrite required to produce 8 kg of H_2SO_4 $= \frac{4896}{80} \times 100$ g (03)

= 6120 g

= 6.12 kg (04)

9b(iii): 25 marks

9(b): 75 marks

10.(a) The following questions are based on the Solvay process.

(i) What is the main product of the Solvay process?

Na_2CO_3 / sodium carbonate (04)

(ii) What is the main by-product of the Solvay process?

$CaCl_2$ / calcium chloride (04)

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(iii) What are the raw materials (starting materials) used in the Solvay process?

NH_3 gas, **(04)**

CO_2 gas, **(04)**

Brine (purified concentrated NaCl solution free from Ca^{2+}, Mg^{2+} and SO_4^{2-} made from sea water **(04)**

(10a(iii): 12 marks)

(iv) Which one of these raw materials in (iii) above is not consumed in the process but is recycled repeatedly?

NH_3 gas **(04)**

(v) Identify the first step of the Solvay process in which raw materials are mixed inside a tower which consists of perforated clay trays. Explain why this is carried out at a low temperature.

Ammonification (mixing NH_3 gas and Brine using the counter current principle) **(02)**

Ammonification is an exothermic process **(03)**

If the temperature increases, efficiency of dissolving NH_3 gas in Brine decreases and hence low temperature is maintained. **(03)**

OR

Dissolution of ammonia is exothermic ($\Delta H < 0$) **(01)**
and the entropy change (ΔS) is negative) ($\Delta S < 0$) **(01)**

According to $\Delta G = \Delta H - T\Delta S$ **(01)**

When ΔS is negative, $-T\Delta S$ term is +ve. **(01)**

When temp increases ΔG becomes less negative, **(01)**

This reduces efficiency of dissolving in water; therefore low Temp more suitable **(03)**

(vi) Give **three** uses of the main product of the Solvay process.

To remove hardness of water

Used in glass industry

Used in paper industry

Used in soap and detergent industry to improve the cleaning action

Used as a powerful cleaning agent (washing soda)

Any three (03 marks x 3 = 09 marks)

(vii) Give **three** reasons contributing to the economic profitability of the Solvay process.

NaCl and $CaCO_3$ can be obtained easily with low cost

NH_3 is not used and can be used repeatedly by recycling

Portion of CO_2 can be reused

(03 marks x 03 = 09 marks)

10(a): 50 marks

(*b*) Briefly explain each of the following statements.

(i) Agriculture contributes to global warming.

Related to agriculture:

- N_2O **(02)** and CH_4 **(02)** contribute to global warming
- Both are greenhouse gases **(02)**
- Increase in their concentration contributes to global warming. **(02)**

Formation of N_2O:

- Action of denitrifying bacteria **(02)** on
- nitrogen containing fertilizers **(02)** produce N_2O

Formation of CH_4:

- Paddy cultivation in marsh and water logged area **(02)** favour anaerobic decomposition of organic matter to produce CH_4. **(01)**
- Decomposition of organic matter disposed indiscriminately **(02)** undergo anaerobic decomposition to produce CH_4. **(01)**
- Digestion of organic/cellulose matter in guts of ruminant animals (cows, goat, sheep) undergoing bacterial decomposition under anaerobic conditions **(02)** produce CH_4

(10b(i): 20 marks)

(ii) Iron extraction contributes to global warming.

Related to iron extraction:

- CO_2 contributes to global warming
- CO_2 is a greenhouse gas and
- increase in concentration contributes to global warming

(02x3)

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Formation of CO_2:

Fossil fuels and coke used in iron extraction are converted to CO_2 during combustion **(04)**

(10b(ii): 10 marks)

(iii) Transportation contributes to photochemical smog.

In your answer indicate how the chemical species responsible for the given environmental effect in each of the statements above is/are formed.

Related to transportation:

Species that contribute to photochemical smog

NO and volatile hydrocarbons ($CH_3(CH_2)_nCH_3$, n = 1-4) **(02x2)**

- NO and volatile hydrocarbons undergo reactions in the presence of sunlight to produce volatile short chain aldehydes and other toxic chemicals (PAN, PBN)
- Aldehydes are further polymerized to produce suspended particles in the atmosphere
- These particles can form large particles by deposition of dust, water vapour, etc
- These large particles can scatter sunlight reducing the visibility/transparency of the atmosphere and appear like a mist in the lower atmosphere

(02 x 8 = 16)

(10b(iii): 20 marks)

10(b): 50 marks

(c) (i) The following questions are based on vinegar production.

I. State what is the process used in the production of natural vinegar.

microbial activity / microbial fermentation **(04)**

II. Write the name of the active chemical ingredient present in natural vinegar.

acetic acid **(04)**

III. Name the titrant and the indicator used in the quantitative determination of the active chemical ingredient in natural vinegar.

Titrant – NaOH **(04)**

Indicator – Phenolphthalein **(04)**

IV. State the difference in composition between natural vinegar and artificial vinegar.

Natural vinegar contains salts, simple sugars, esters and alcohols in minute quantities. **(01x4)**

However, synthetic vinegar contains only acetic acid. **(02)**

(ii) The following questions are based on the extraction of essential oils from plants.

I. Name **three** methods that can be used to extract essential oils.

steam distillation

solvent extraction

pressing

(04 marks x 3 = 12 marks)

II. State which of the above methods is based on the application of Daltons Law of partial pressures.

steam distillation

(04 marks)

III. Name the major compound present in each of the essential oils given below.

- Citronella oil
- Cinnamon root oil
- Cinnamon leaf oil

citronella – geraniol

cinnamon root – camphor

cinnamon leaf - eugenol

(04 marks x 3 = 12 marks)

10(c): 50 marks

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